C PROGRAMMING: POWERING EVERYDAY TECHNOLOGY

Presented By: Aryan Chandra
Mihir Pingle
Tanish Venkatesh
Bhaumik Ghode
Yash Gupta

01

INTRODUCTION TO C

C IS THE BACKBONE OF MANY TECHNOLOGIES WE USE EVERY DAY.

- C is a powerful and efficient programming language developed for system and hardware-level programming
- It works very close to the computer's hardware, giving high speed and control
- C is widely used in embedded systems and real-time applications
- Many everyday technologies like ATMs, traffic lights, elevators, and home appliances are powered by C
- Its reliability and performance make it ideal for machines that must run continuously

IMPORTANCE OF C

02

ESSENTIAL FOR PERFORMANCE AND CONTROL

DIRECT CONTROL

C provides direct control over memory and hardware, allowing programmers to optimize performance for specific applications.

SPEED

Programs written in C run very quickly, making it suitable for applications that require fast processing and response times.

REAL-TIME SYSTEMS

C is ideal for real-time systems, where timing and accuracy are critical for tasks like monitoring and control.

SMART AIR CONDITIONERS SYSTEM

Why C Language?

- Fast execution
- Low memory usage
- Direct communication with hardware
- Reliable for real-time embedded systems

Application

C programming enables the air conditioner to automatically monitor temperature, control motors, and operate safely without human intervention.

03

REAL-LIFE APPLICATIONS

A ATM Machines

B Traffic Light Control System

C Washing Machines

D Elevator Systems

E Embedded Systems (Microcontrollers)

F Operating Systems

05

HOW THESE APPLICATIONS REALLY WORK

```
int pin = 1234, enteredPin, attempts = 0;

while(attempts < 3) {
  printf("Enter PIN: ");
  scanf("%d", &enteredPin);

  if(enteredPin == pin) {
    printf("Access Granted\n");
    break;
  } else {
    printf("Wrong PIN\n");
    attempts++;
  }
}
```

ATM Machine (Security & Transactions)

An ATM uses C programming to handle secure user interactions. When a user enters a PIN, the program uses if-else conditions to check whether the PIN is correct. A while loop limits the number of attempts to prevent misuse. C also communicates directly with hardware components like the keypad, card reader, display screen, and cash dispenser. Once verification is successful, the program allows transactions such as withdrawal or balance enquiry.

Traffic Light Control System

Traffic lights use C programs running in an infinite loop. Timers and delays are used to decide how long each light stays on. For example, red stays on for a fixed time, then switches to green, then yellow. The loop ensures this cycle runs continuously without stopping. Since traffic systems must work 24/7, C is used because it is fast, reliable, and suitable for real-time control.

```
while(1) {
  printf("RED Light ON\n");
  delay(5000);

  printf("GREEN Light ON\n");
  delay(5000);

  printf("YELLOW Light ON\n");
  delay(2000);
}
```

06

```
int mode;

printf("Select Wash Mode (1-Quick, 2-Normal, 3-Heavy): ");
scanf("%d", &mode);

switch(mode) {
   case 1: printf("Quick Wash\n"); break;
   case 2: printf("Normal Wash\n"); break;
   case 3: printf("Heavy Wash\n"); break;
   default: printf("Invalid Mode\n");
}
```

Washing Machine (Appliance Control)

In washing machines, C is used inside a microcontroller. The program reads data from sensors such as water level and temperature. Using switch-case statements, the machine selects different wash modes like quick wash or heavy wash. Motors, valves, and heaters are controlled based on sensor input. C ensures accurate timing and smooth operation of the washing process.

Elevator System (Real-Time Monitoring)

Elevators use C programming to manage **movement** between floors safely. When a button is pressed, a switch-case decides the target floor. The code loops continuously to monitor sensors for door position, floor alignment, and load. **Safety** checks run in real time to prevent accidents. C is used because it gives precise control and quick response.

```
int floor;

printf("Enter floor number: ");
scanf("%d", &floor);

switch(floor) {
   case 1: printf("Going to Floor 1\n"); break;
   case 2: printf("Going to Floor 2\n"); break;
   case 3: printf("Going to Floor 3\n"); break;
   default: printf("Invalid Floor\n");
}
```

07

```
int sensorValue = 1;

if(sensorValue == 1) {
   printf("Device ON\n");
} else {
   printf("Device OFF\n");
}
```

Embedded Systems (Microcontrollers)

Embedded systems like routers, smart meters, and IoT devices use C because it runs efficiently on limited hardware. C programs directly control memory, input/output pins, and sensors. The program continuously checks inputs and responds immediately, making it ideal for small devices.

Operating Systems (System Software)

Operating systems use C to manage memory, processes, and hardware resources. C allows direct interaction with the CPU and memory while still being portable. This makes it possible to build fast and stable system software.

```
while(1) {
   printf("Running System Task\n");
}
```

THANK YOU